Single-channel EEG seizure detection on CHB-MIT: accuracy, model size and efficiency

Bold = best in column. <u>Underlined</u> = second best. Higher is better for ACC and SEN; lower is better for parameters, stored footprint and MACs. Every row is CHB-MIT. Footprints marked *est.* are derived from a parameter count rather than reported by the paper, and are excluded from the ranking. EpiSepNet-5K is this group's own earlier model, kept in the table as the design it succeeds. Weight memory and total on-chip memory are different quantities: most papers report only the first. **DFP** is dynamic fixed point, a power-of-two scale: its footprint EQUALS plain integer at the same width, because the scale is per tensor rather than per value. What it changes is the datapath — requantising between layers becomes a shift instead of a multiply.

Method	Dataset	Model	Protocol / test exposure	ACC (%)	SEN (%)	Parameters	Precision	Weight memory	Total on-chip (W+A)	Other efficiency
WearSeizure-1D (this work)	CHB-MIT (13 single-channel cases)	Separable 1D-CNN, k5 + dilation 1/2/4/8/16; 1 channel; 4-s windows	**Leakage-safe**: split by recording before filtering; thresholds frozen on val; **185.0 h** continuous test; 66 folds x 3 seeds	98.88	94.89 event level; 60.33 segment level	11,786	INT8/DFP8 or **INT16/DFP16** format not yet fixed; being chosen by measured loss	**11.5 KB** at INT8 or DFP8 **23.0 KB** at INT16 or DFP16	**18.2 KB** at INT8/DFP8 **36.3 KB** at INT16/DFP16 72.7 KB at FP32 weights + line buffers, peak measured per layer	<u>585,920 MACs (thop) / 489,600 conv+fc; 6.7 KB line buffers; FAR 0.29/h; delay 17.8 s</u>
WearSeizure-1D, same code under the segment-split protocol	CHB-MIT (13 single-channel cases)	As above, retrained under the leaky protocol	*Segment-level random split*; normalised on all data; threshold on test — i.e. the protocol common to the rows below. 99.6% of test windows have a near-duplicate in training	**99.68**	92.29 segment level	11,786	FP32 (measured)	as above	as above	Same network; only the split rule differs from the row above
EpiSepNet-5K (INT16) earlier model, same group	CHB-MIT	BatchNorm-folded separable 1D-CNN	NR	90.04	90.76	**4,900**	INT16	<u>10.0 KB package; ~9.8 KB raw INT16</u>	NR	99.9743% agreement with FP32; 2.81x smaller package
EpiSepNet-5K (FP32) earlier model, same group	CHB-MIT	Separable 1D-CNN; 17-channel raw EEG; 2-s windows	NR	90.07	90.76	<u>5,010</u>	FP32	28.1 KB checkpoint; ~20.0 KB raw FP32	NR	Reference model
Chung et al. 2024 [1]	CHB-MIT (13 cases)	Stacked 2D-CNN, parallel 1x3 and 1x5 kernels; 1 clinician-selected channel	Segment-level 70/20/10 split per patient; ~91 h	98.18	**99.62** event level; 96.76 segment level	116,700	FP32	**454 KB** at FP32 rebuilt from the paper; 116 226 params vs 116 700 reported, +0.4%	**510 KB** at FP32 = **91% of XC7Z020 BRAM** 128 KB at INT8 (23%) peak 14 464 values at stage 2, both branches live	FAR 0.22/h; detection delay 3.3 s; specificity 98.19%
Werner et al. (TC-ResNet4) [2]	CHB-MIT	16-channel TC-ResNet4 with fixed-point inference	NR	95.28	92.34	9,840	4-bit	**~4.92 KB weight-only (4-bit)**	NR	**337,968 MACs; 495 nW average power**
Zhu et al. 2021 [3]	CHB-MIT (23 channels)	7-layer 1D-CNN, 5x1 conv, shift-register PE array	80/10/10 split; normal class down-sampled 200:1	97.35	94.32	7,010	fixed-point	NR; *est.* 28 KB at FP32, 7 KB at 8-bit	NR	6.32 MOPs; 170 us/inference @ 200 MHz; FPGA Xilinx Zynq ZC706
Li et al. 2022 [4]	CHB-MIT + Bonn + SWEC-ETHZ	1D-CNN with parallel convolutional layers	80/20 + 5-fold CV; GAN-synthesised preictal segments	<u>99.01</u>	<u>99.24</u>	10,778	analog RRAM	NR; *est.* 43 KB at FP32	weights held in RRAM crossbar, not SRAM	1.13 us parallelised; 7.21 W; ASIC 22 nm FDSOI + RRAM crossbar
Ferrara et al. [5]	CHB-MIT	Patient-specific two-channel lightweight CNN	NR	99.00	67.00	9,500	NR	51 KB model	NR	Balanced accuracy 83.0%; 0.10 FP/h
REST-RS [6]	CHB-MIT	Graph-based residual state-update model	NR	NR	NR	9,300	NR	0.037 MB, ~37 KB	NR	AUROC up to 93.5%; 1.314 ms inference
SlimSeiz [7]	CHB-MIT	Eight-channel convolution and Mamba network (prediction task)	NR	94.80	95.50	21,200	NR	NR; *est.* 84.8 KB at FP32	NR	Specificity 94.0%
RGF-Model [8]	CHB-MIT	Multi-teacher knowledge-distillation model	NR	98.92	98.54	82,000	FP32	0.33 MB (330 KB)	NR	Specificity 99.11%; AUC 98.96%
Wang et al. (MSCA) [9]	CHB-MIT	Inverted residual CNN with multi-scale channel attention	NR	98.70	98.30	88,000	NR	NR; *est.* 352 KB at FP32	NR	2.68M MACs; specificity 99.10%
Ahlawat (INT8) [10]	CHB-MIT	Quantized common-channel 1D-CNN with operator fusion	NR	NR	NR	NR	INT8	0.44 MB (440 KB) at INT8; 1.63 MB at FP32	NR	Up to 2.8x speedup; up to 64% estimated energy reduction

Reading the table. Protocols differ between rows, and most papers do not state theirs in enough detail to compare. This work is the only row evaluated with splits taken by recording, thresholds frozen on a held-out validation partition, and 185 h of continuous test exposure. Under the segment-level random split used by much of this literature, the same code and data reach 92.29 % segment sensitivity instead of 60.33 % — a 31-point difference caused by the protocol alone, because 99.6 % of test windows then have a near-duplicate in training. Accuracy is nearly blind to this: at 0.62 % ictal prevalence a model that never predicts a seizure already scores 99.38 %.

References

[1] Y. G. Chung, A. Cho, H. Kim, K. J. Kim. Single-channel seizure detection with clinical confirmation of seizure locations using CHB-MIT dataset. Front. Neurol. 15:1389731, 2024.
[2] J. Werner, B. Kohli, P. Palomero Bernardo, C. Gerum et al. TC-ResNet for EEG seizure detection.
[3] L. Zhu, D. Liu, X. Li, J. Lu, L. Wei, X. Cheng. An Efficient Hardware Architecture for Epileptic Seizure Detection using EEG Signals based on 1D-CNN. IEEE ASICON, 2021.
[4] C. Li, C. Lammie, X. Dong, A. Amirsoleimani, M. Rahimi Azghadi, R. Genov. Seizure Detection and Prediction by Parallel Memristive Convolutional Neural Networks. IEEE TBioCAS, 2022.
[5] Ferrara et al. Patient-specific two-channel lightweight CNN for seizure detection.
[6] REST-RS: graph-based residual state-update model for EEG seizure detection.
[7] SlimSeiz: convolution and Mamba network for seizure prediction.
[8] RGF-Model: multi-teacher knowledge distillation for seizure detection.
[9] Wang et al. Lightweight Seizure Detection Based on Multi-Scale Channel Attention, 2023.
[10] K. Ahlawat. Efficient EEG Seizure Detection Using INT8 Quantization, Channel Pruning, and Spiking Neural Networks. arXiv:2607.16296, 2026.